Medicare ESRD PPS Pricer - Core Business Rules

The following business rules govern the pricing calculations for the End-Stage Renal Disease (ESRD) Prospective Payment System (PPS) for Calendar Year 2021 (Version 21.2).

Rule 1: Geographic Wage Index Lookup & Application

Description: The base payment rate for a dialysis treatment is split into a "labor" portion and a "non-labor" portion. The labor portion is adjusted based on the cost of living in the facility's specific geographic area (CBSA).

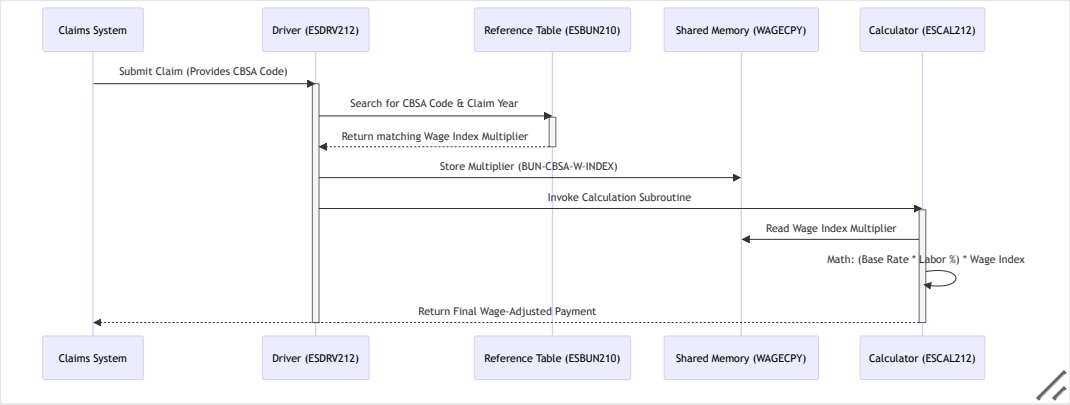
```
Rule: Apply Geographic Wage Index to Labor Costs
  Given a facility submits a claim containing their 5-digit Core-Based Statistical Area (CBSA) code
  And the claim contains a valid Date of Service
  When the routing system searches the Bundled Wage Index Reference Table
  And matches the facility's CBSA code and the specific Year of the claim
  Then the system retrieves the designated Wage Index Multiplier for that region
  And transfers this multiplier into shared memory
  And the calculator multiplies the National Base Payment Rate by the National Labor Percentage
  And multiplies that result by the retrieved Wage Index Multiplier
  And adds the unadjusted Non-Labor portion to yield the final Wage-Adjusted Base Rate
```

Code Path Citations

- **`BILLCPY.txt` (Line 005300):** Defines `P-GEO-CBSA`. This is where FISS passes the facility's geographic code.
- **`ESDRV212.txt` (Lines 107800 - 109100):** The Driver executes a binary `SEARCH ALL` against the loaded reference table.

- `ESBUN210.txt` **(Lines 009400, 020100 - 020500):** The hardcoded Reference Data table mapping CBSA codes to memory pointers.
- `WAGECPY.txt` **(Line 004300):** Defines `BUN-CBSA-W-INDEX`, the shared memory bridge passing the multiplier to the Calculator.
- `ESCAL212.txt` **(Lines 081900 - 082600):** Applies the business math: `(Base Rate * Labor %) * Wage Index`.

Sequence Diagram



Rule 2: Children's Hospital Wage Override

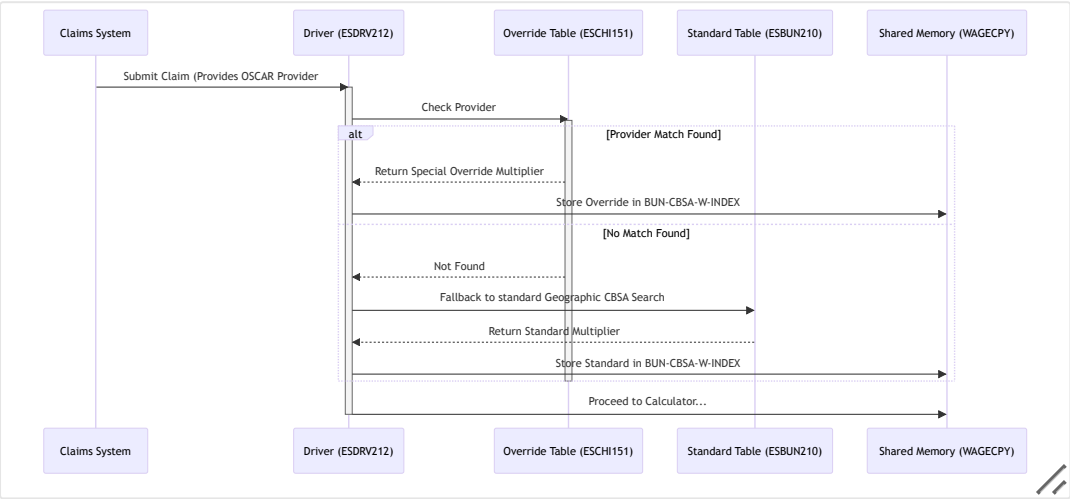
Description: Designated Children's Hospitals are exempt from standard geographic CBSA lookups. They receive a predetermined wage index based entirely on their unique facility provider number.

```
Rule: Children's Hospital Special Wage Override
  Given a facility submits a claim containing their 6-
character OSCAR Provider Number
  And the claim's Date of Service falls within
Calendar Year 2015 or later
  When the routing system checks the Special
Children's Hospital Reference Table
  And finds a direct match for the submitted Provider
Number
  Then the system bypasses the standard geographic
CBSA lookup entirely
  And retrieves the specialized Wage Index Multiplier
mapped to that specific hospital
  And transfers this override multiplier into shared
memory
  And the calculator applies this override to the
labor portion of the payment
```

Code Path Citations

- **`BILLCPY.txt` (Line 005100):** Defines `P-PROV-OSCAR`.
- **`ESDRV212.txt` (Lines 105100 - 106100):** Checks the claim date and forces a search against the Children's Hospital table (`0820-SEARCH-CHILD-HOSP-TABLE`).
- **`ESCHI151.txt` (Lines 001320 - 001700):** The hardcoded Reference table for overrides (e.g., Provider `093300` receives `1.0415`).

Sequence Diagram



Rule 3: Supplemental Wage Index 5% Cap

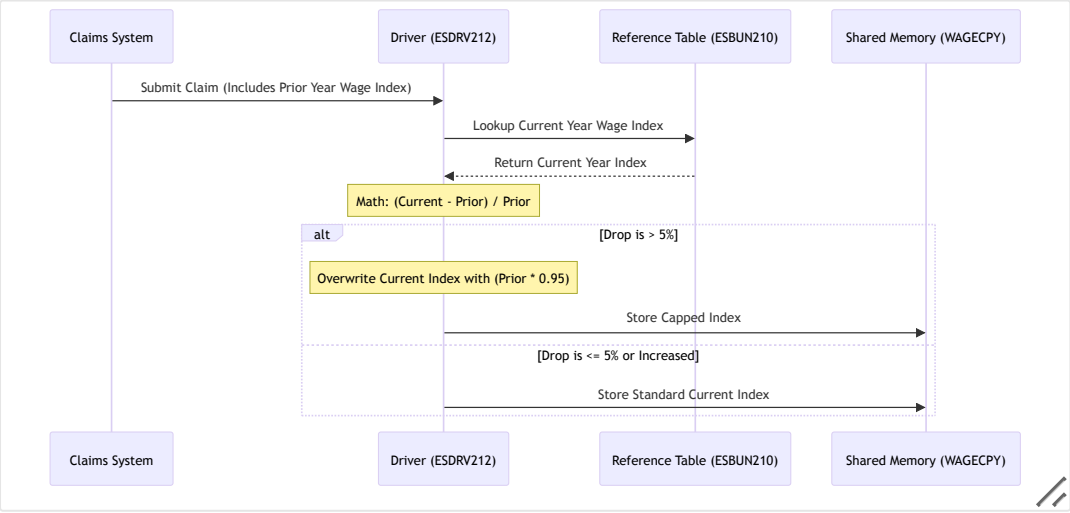
Description: To prevent facilities from experiencing severe financial shocks, a facility's wage index cannot drop by more than 5% from the previous year.

```
Rule: Supplemental Wage Index Decrease Cap
  Given a claim with a Date of Service after September
30, 2020
  And the Provider Specific File has the "Supplemental
Wage Index" indicator turned ON ('1')
  And the system is provided the facility's exact Wage
Index from the prior year
  When the system retrieves the current year's
standard Wage Index based on geography
  And calculates the percentage change: (Current Index
- Prior Index) / Prior Index
  If the calculated percentage represents a drop
greater than 5 percent (< -0.05)
  Then the system rejects the current year's standard
index
  And recalculates a mitigated index set to exactly 95
percent of the prior year's value
  And transfers this mitigated index to the calculator
```

Code Path Citations

- **`BILLCPY.txt` (Lines 006500 - 006600):** Defines `P-SUPP-WI-IND` (ON/OFF switch) and `P-SUPP-WI` (Prior year value).
- **`ESDRV212.txt` (Lines 109800 - 111300):** Checks if ratio drops below `-0.05`. Overwrites variable with `P-SUPP-WI * 0.95`.

Sequence Diagram



Rule 4: Acute Kidney Injury (AKI) Exemption

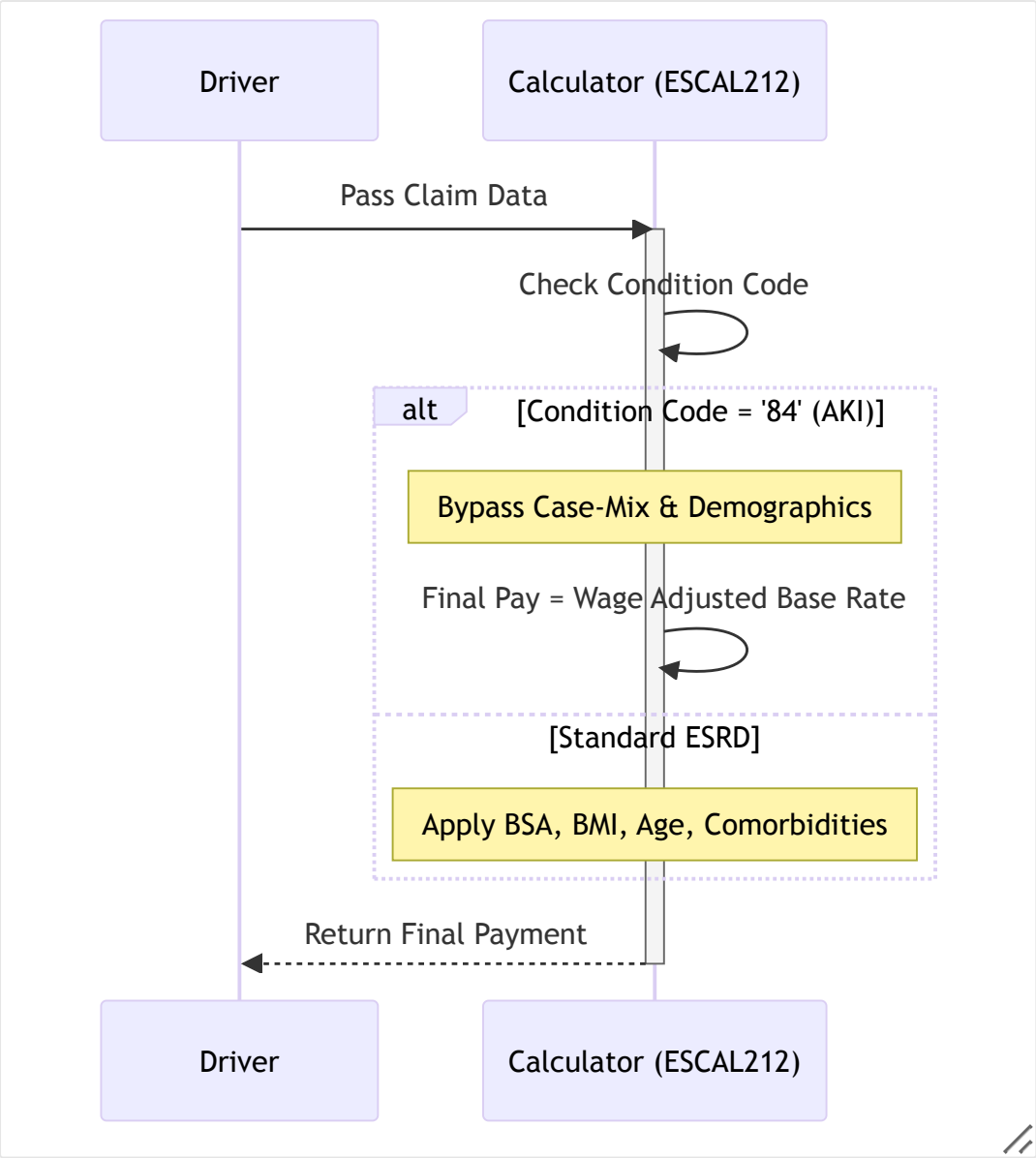
Description: Claims for Acute Kidney Injury are priced using a simplified baseline structure and are exempt from standard ESRD demographic adjustments and quality penalties.

```
Rule: AKI Pricing Bypass
  Given an incoming dialysis claim
  When the Condition Code is '84' (Acute Kidney
Injury)
  Then bypass validation edits for Patient Height and
Weight
  And bypass all Case-Mix adjustments (Age, BMI, BSA,
Comorbidities)
  And do not apply the Quality Incentive Program (QIP)
payment reduction
  And set the Final Payment Amount directly to the
Wage-Adjusted Base Rate
```

Code Path Citations

- **`ESDRV212.txt` (Lines 069300 - 071600):** `1000-VALIDATE-BILL-ELEMENTS` bypasses Height/Weight edits if Condition Code is 84.
- **`ESCAL212.txt` (Lines 065800 - 066300):** In `0000-START-TO-FINISH`, detects CC '84', skips `2000-CALCULATE-BUNDLED-FACTORS`, and sets final payment directly to `H-BUN-BASE-WAGE-AMT`.

Sequence Diagram



Rule 5: Pediatric Age Classification

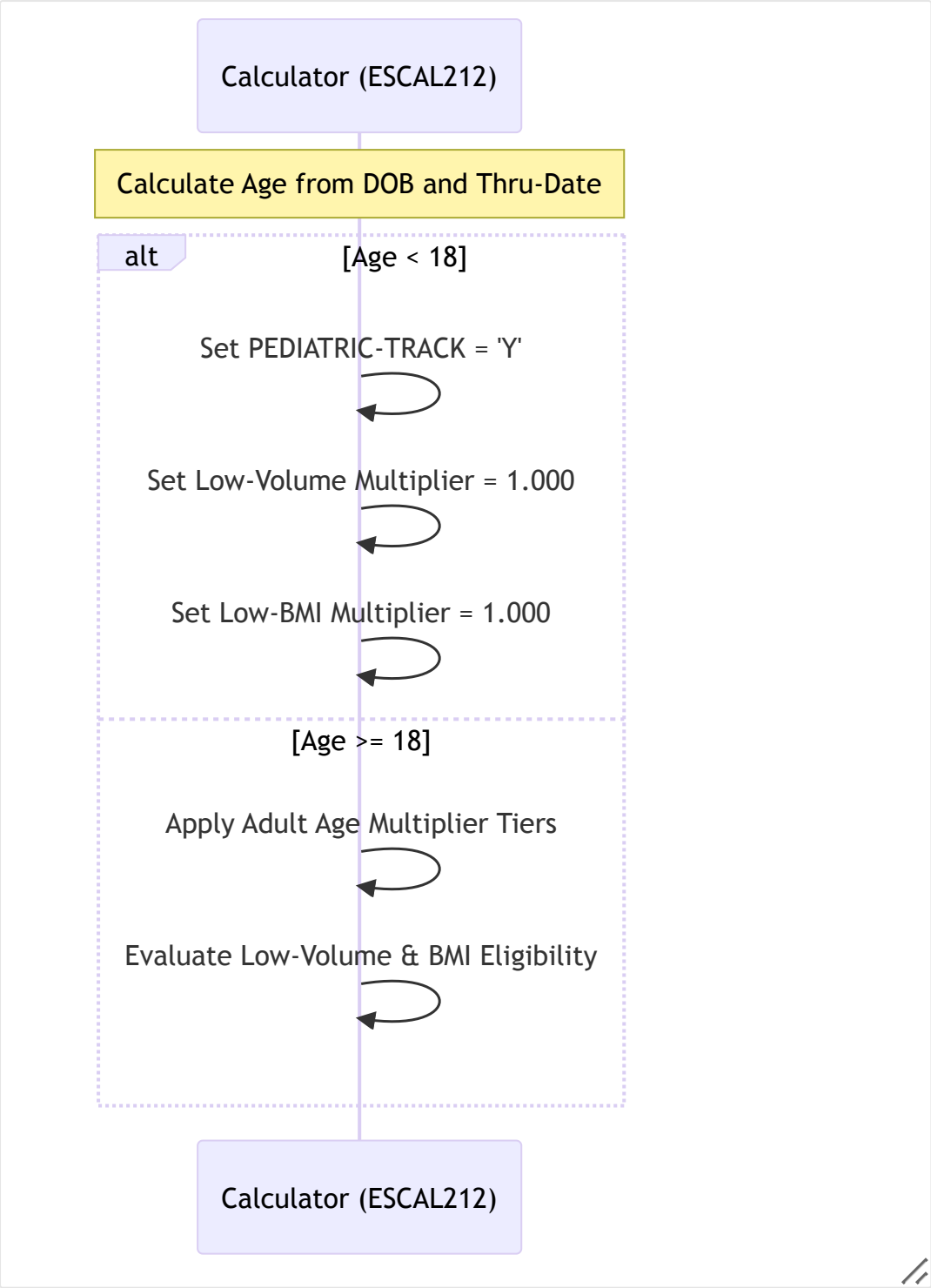
Description: Defines the exact threshold for classifying a dialysis patient as pediatric, altering base multipliers and preventing adult-only adjustments (like Low-Volume).

```
Rule: Pediatric Classification Threshold
  Given a patient claim with a valid Date of Birth and Through Date
  When the patient's age is calculated (Through Year - Birth Year, adjusted for Birth Month)
  And the calculated age is strictly less than 18 years
  Then classify the claim as Pediatric
  And set the internal "PEDIATRIC-TRACK" indicator to "Y"
  And bypass the adult Low-Volume and Low-BMI multipliers
  And apply Pediatric-specific Age Multipliers based on Modality
```

Code Path Citations

- **`ESCAL212.txt` (Lines 083000 - 083600):** `2000-CALCULATE-BUNDLED-FACTORS` determines age. If `< 18`, sets `PEDIATRIC-TRACK` to "Y".
- **`ESCAL212.txt` (Lines 110000 - 110400):** Forces Low-Volume multiplier to `1.000` (neutral) if Age `< 18`.

Sequence Diagram



Rule 6: Low Body Mass Index (BMI) Adjustment

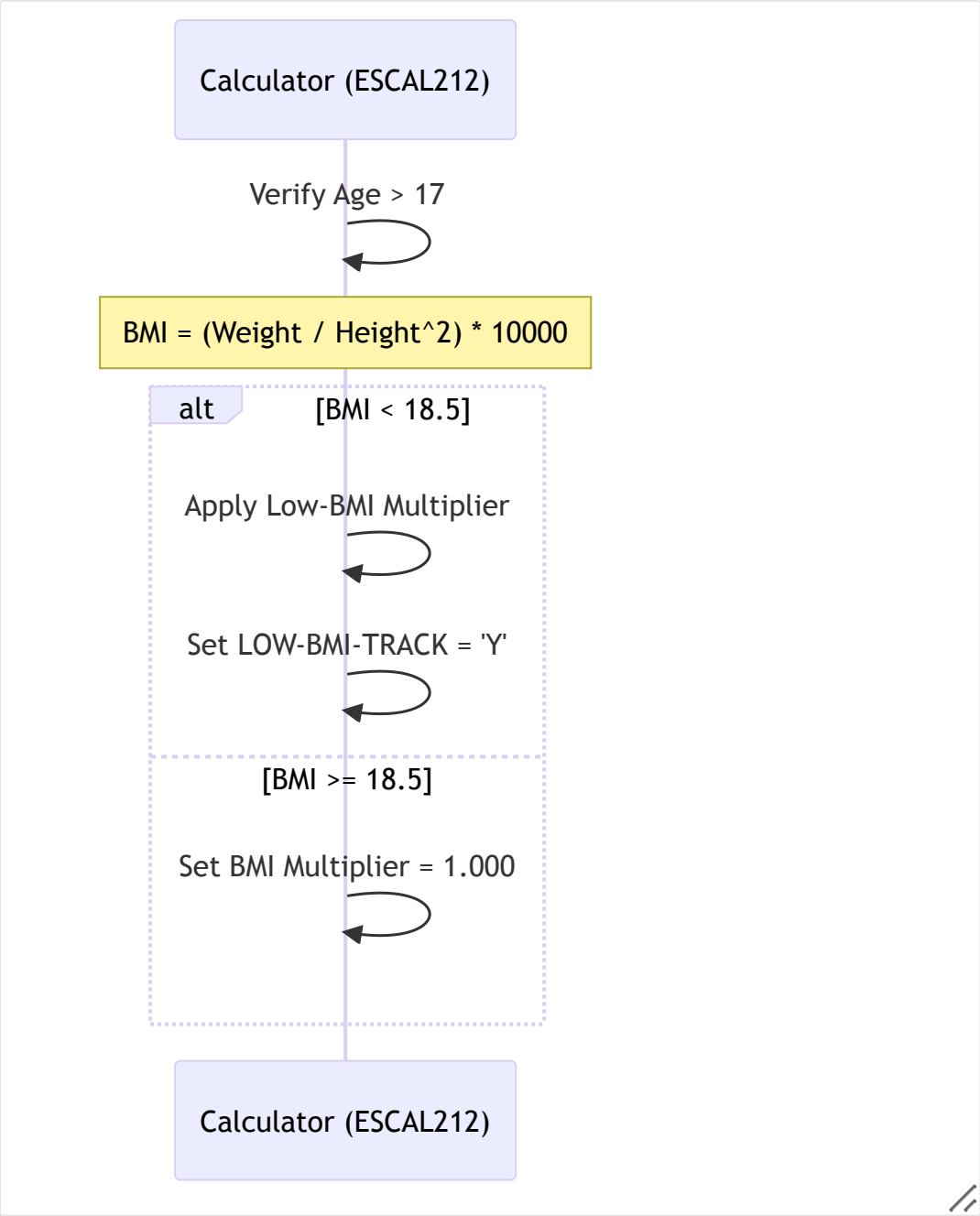
Description: Adult patients with an abnormally low BMI receive a specific payment multiplier to account for the increased complexity of care.

```
Rule: Low BMI Case-Mix Adjustment
  Given the patient is classified as an Adult (Age >
17)
  When the patient's BMI is calculated as: (Weight /
Height^2) * 10000
  And the calculated BMI is strictly less than 18.5
  Then apply the 'CM-BMI-LT-18-5' multiplier to the
Base Wage Amount
  And trigger the internal "LOW-BMI-TRACK" indicator
```

Code Path Citations

- **`ESCAL212.txt` (Lines 100600 - 101400):** Executes the `(Weight / (Height ** 2)) * 10000` formula. If `H-BUN-BMI < 18.5`, assigns `CM-BMI-LT-18-5` to `H-BUN-BMI-FACTOR`.

Sequence Diagram



Rule 7: Dialysis Onset (New Patient) Adjustment

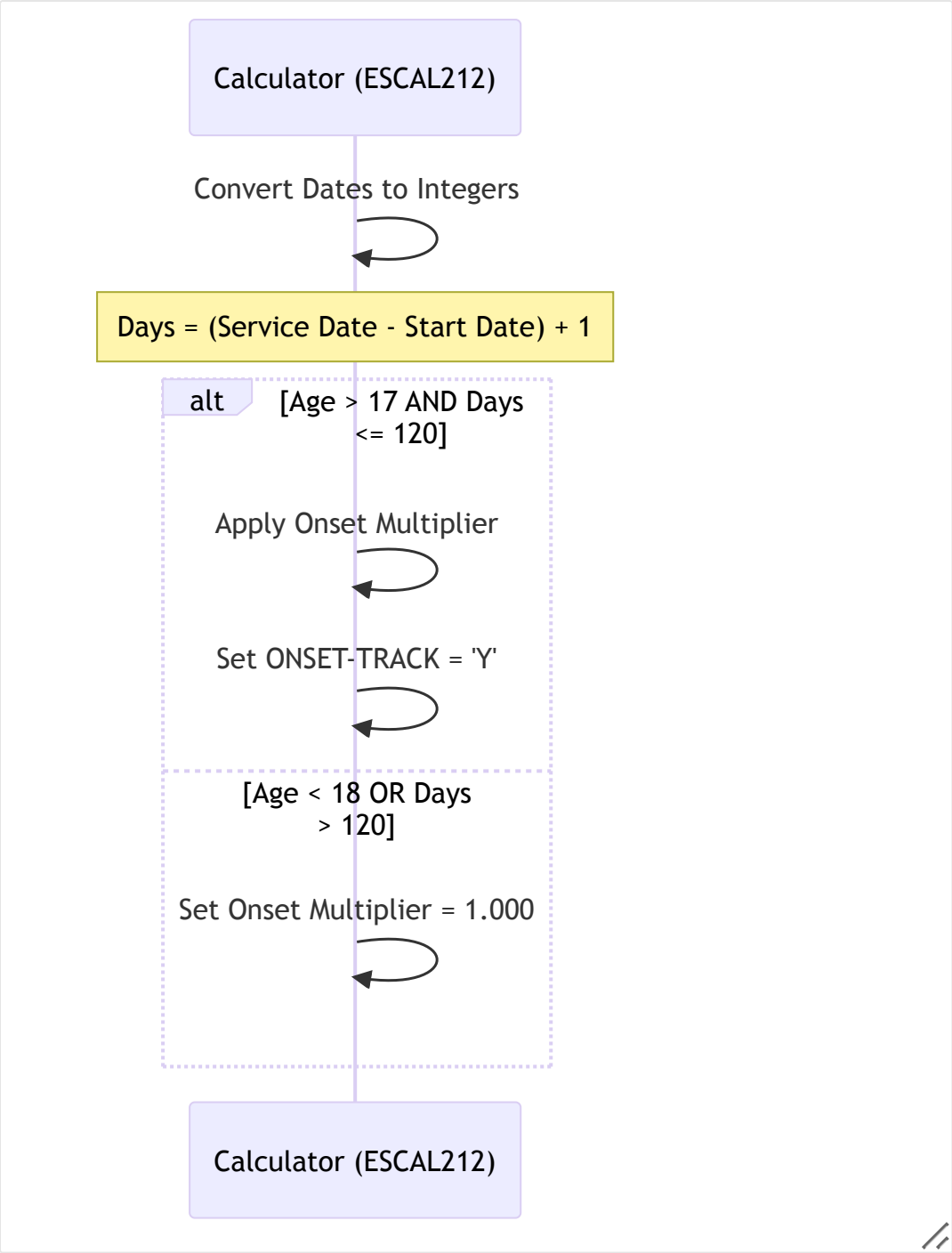
Description: Patients who are new to dialysis receive a temporary adjustment multiplier, provided they have been on dialysis for 120 days or less.

```
Rule: 120-Day Onset Multiplier
  Given the claim contains a valid Dialysis Start Date
  And the patient is classified as an Adult (Age > 17)
  When the duration between the Dialysis Start Date
and the Line-Item Date of Service is calculated
  And the duration is 120 days or less (inclusive of
the start date)
  Then apply the 'CM-ONSET-LE-120' multiplier to the
Base Wage Amount
  And trigger the internal "ONSET-TRACK" indicator
```

Code Path Citations

- **`ESCAL212.txt` (Lines 101900 - 104200):** Converts dates to integers via `FUNCTION INTEGER-OF-DATE`. Calculates `ONSET-DATE = (LINE-ITEM-DATE - DIALYSIS-DATE) + 1`. If `<= 120`, applies `CM-ONSET-LE-120`.

Sequence Diagram



Rule 8: Quality Incentive Program (QIP) Reductions

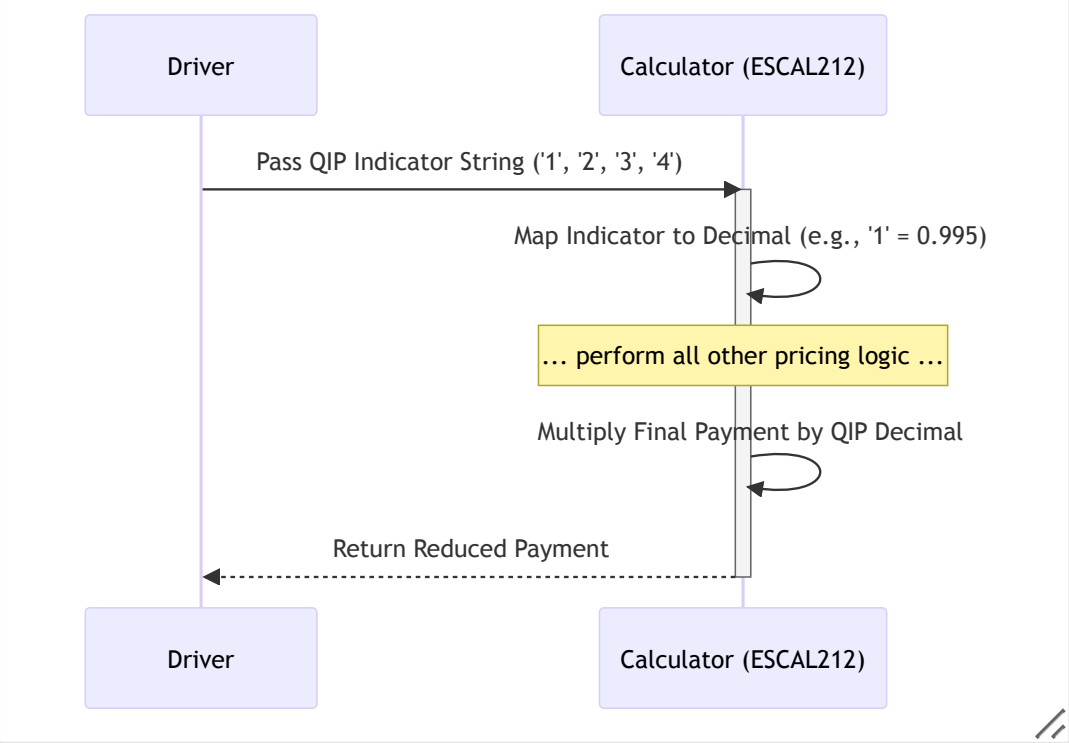
Description: Facilities that fail to meet quality care metrics receive a tiered penalty reduction on their final payment.

```
Rule: QIP Penalty Tiers
  Given the Provider Specific File contains a QIP
Reduction indicator
  And the claim does NOT contain Condition Code '84'
(AKI)
  When the QIP Indicator evaluates to:
    | Indicator | Then Apply Payment Multiplier |
    | '1'       | 0.995 (0.5% Reduction)        |
    | '2'       | 0.990 (1.0% Reduction)        |
    | '3'       | 0.985 (1.5% Reduction)        |
    | '4'       | 0.980 (2.0% Reduction)        |
  Then multiply the Final PPS Payment Rate by the
selected multiplier
```

Code Path Citations

- **`ESCAL212.txt` (Lines 084000 - 086100):** Translates the string indicator (e.g., '1') into the decimal multiplier (e.g., `0.995`).
- **`ESCAL212.txt` (Lines 200000 - 201800):** Applies the `QIP-REDUCTION` to the `PPS-2011-FULL-PPS-RATE` variable at the end of the program.

Sequence Diagram



Rule 9: Comorbidity Selection Priority

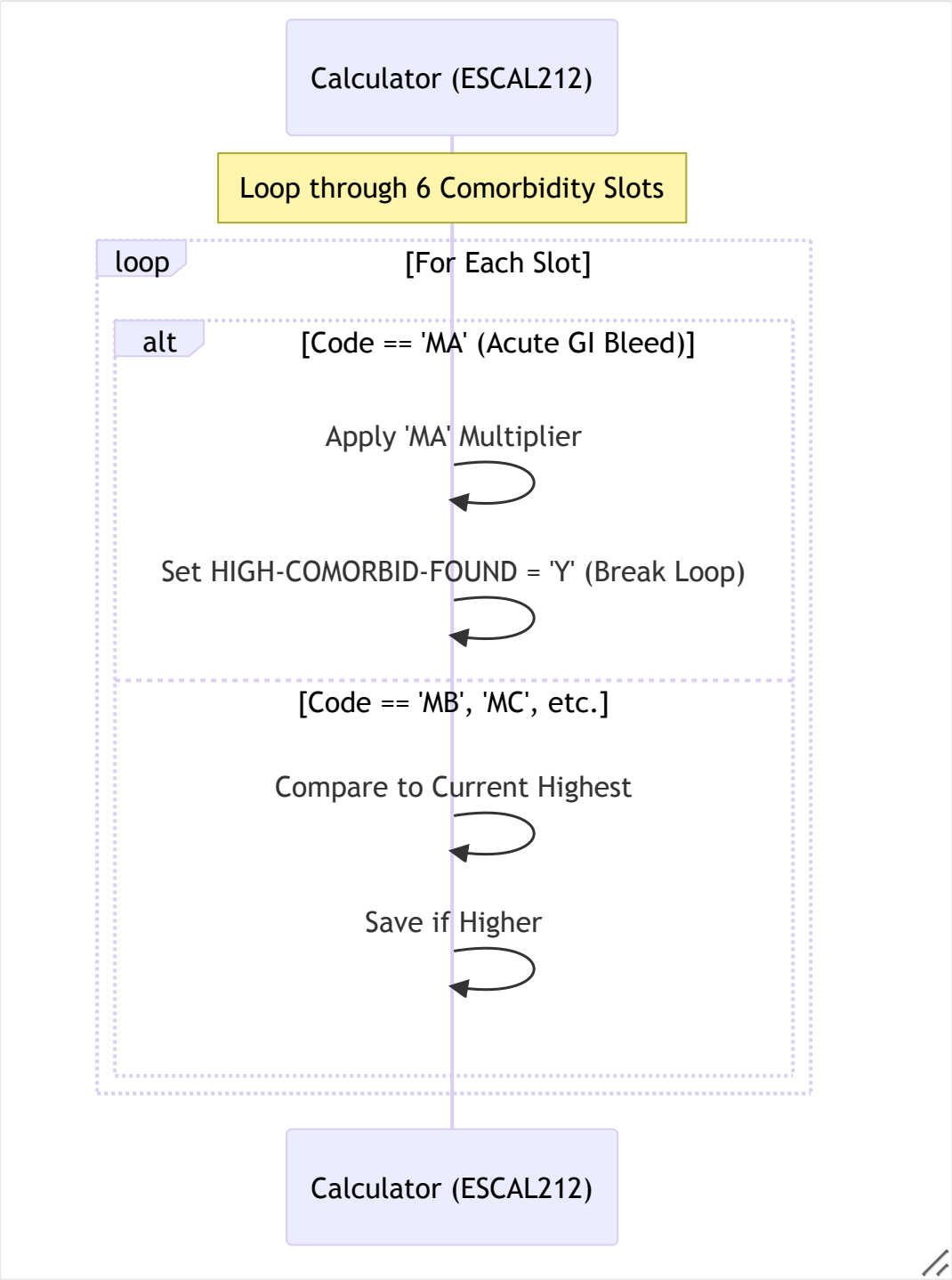
Description: A patient may have multiple comorbidities listed on a claim, but Medicare only pays for one. The system prioritizes acute conditions over chronic conditions.

```
Rule: Comorbidity Single-Selection Hierarchy
  Given a claim contains multiple valid Comorbidity
Codes
  When evaluating which Case-Mix multiplier to apply
  Then select only the single highest-paying
multiplier
  And prioritize acute identifiers (e.g., 'MA' GI
Bleed)
  Over chronic identifiers (e.g., 'MD'
Myelodysplastic)
  And apply the single selected multiplier to the Base
Wage Amount
```

Code Path Citations

- **`ESCAL212.txt` (Lines 123200 - 127000):** `2100-CALC-COMORBID-ADJUST` loops `UNTIL SUB > 6 OR HIGH-COMORBID-FOUND`. If 'MA' is found, it immediately sets `HIGH-COMORBID-FOUND` to True and exits the loop.

Sequence Diagram

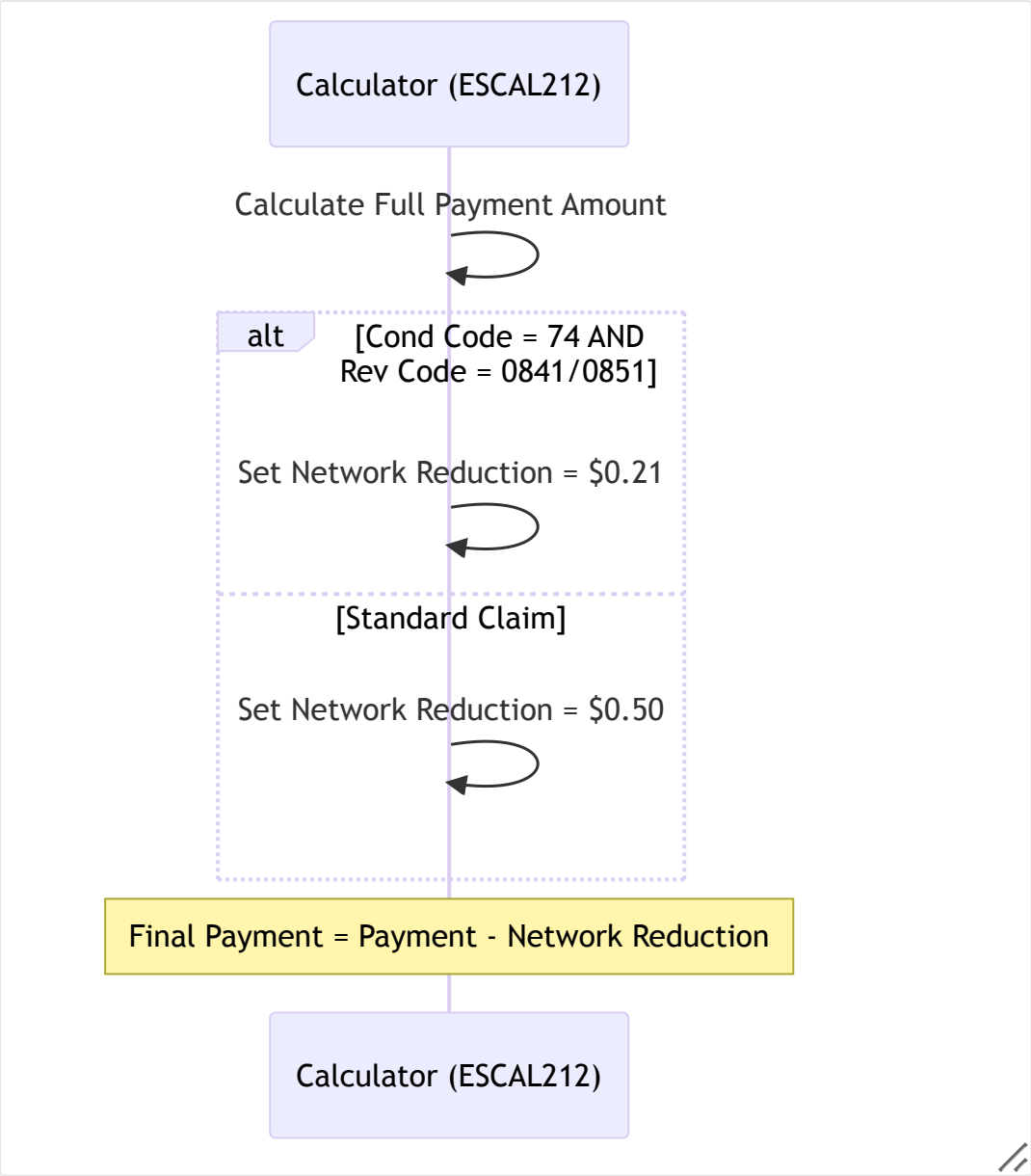


Rule 10: ESRD Network Fee Reduction

Description: A mandated statutory reduction is taken from every dialysis payment to fund the ESRD Network organizations, varying slightly based on how the dialysis is billed.

```
Rule: Network Fee Deduction
  Given a standard ESRD claim (Not AKI)
  When the claim is billed as Home Dialysis (Condition
Code '74')
  And the Revenue Code is '0841' or '0851' (Per-Diem
billing)
  Then deduct exactly $0.21 from the Final Payment
Amount
  Otherwise, for standard full PPS claims
  Then deduct exactly $0.50 from the Final Payment
Amount
```

Code Path Citations

- **`ESCAL212.txt` (Lines 134700 - 135000):** Checks `B-COND-CODE = '74'` and `B-REV-CODE = '0841' OR '0851'`. If true, sets `H-NETWORK-REDUCTION` to 0.21, else 0.50.
- **`ESCAL212.txt` (Lines 216400 - 216700):** Subtracts `H-NETWORK-REDUCTION` from the final calculated amount just before returning.

Sequence Diagram